Project report — Cognitive Growth Agent

1. Introduction

Cognitive Growth Agent is a cognitive training web application for children, developed for the CISTAC 1.0 – AI4Brain challenge. It adapts its activities to four profiles: typical development, ADHD, autism and dyslexia.

Children with ADHD, autism or dyslexia have different learning needs. One needs short series and breaks to move, another needs predictable routines and ways to communicate without speaking, another needs help with reading and writing. Mainstream apps often give every child the same exercises, with timers and rankings.

Our application starts from a profile chosen by the parent, then adjusts the content, the pace and the help to each child. It measures progress with a visible uncertainty, explains each of its choices and stays strictly non-clinical: it never makes a diagnosis.

2. Objectives

The main objective is to help each child progress at their own pace, with activities adapted to their profile and level of need, without ever assessing or labelling their health.

Specific objectives:

1. **Assess** five cognitive dimensions (working memory, attention, processing speed, logical reasoning, mental flexibility) with an estimate and a 95% confidence interval.
2. **Adapt** the activities to the profile (ADHD, autism, dyslexia) and degree (mild, moderate, severe) chosen by the parent.
3. **Help the child progress** with a difficulty that follows their results: 3 right answers move up one level, 2 mistakes out of 3 move down one level.
4. **Vary** the questions: a child sees a question again only after seeing all the others at their level.
5. **Explain** every recommendation with numbers, through a coach that only cites the child's own data.
6. **Measure** collaboration with AI: does the child check what the coach says before trusting it?
7. **Protect** data: consent, local storage, automatic deletion after 90 days.

3. Target users and ethical principles

The application is for children aged 5 to 17, with a parent or guardian; an adult can also use it for themselves. It is not a medical device, a screening tool or a therapy.

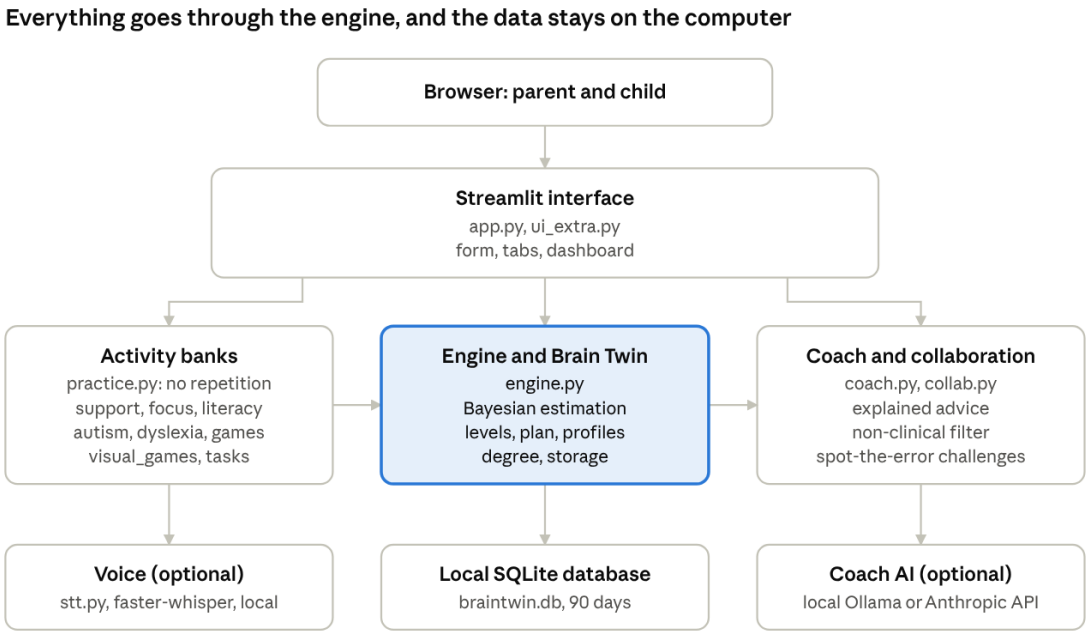
Principle	How it is applied
Non-clinical	The profile and degree are chosen by the parent, never guessed from the answers. The coach filters medical vocabulary.

Principle	How it is applied
Consent	A required checkbox before anything is stored; first name or nickname only, no surname.
Local data	SQLite database on the computer; nothing is sent to a server, unless the parent turns on the Anthropic API for the coach.
Retention	Automatic deletion after 90 days; Delete and Export (JSON) buttons in the sidebar.
Voice	The recording is transcribed in memory, on the computer, then discarded; typing always works.
Kindness	No ranking, no timer in calm mode, positive messages, several right answers accepted when it makes sense.

The `SAFE_LABELS=1` option replaces the profile names with neutral labels (for example “Reading support” instead of “Dyslexia”); the behaviour stays the same.

4. Technical architecture

The application is written in Python 3.11, with Streamlit for the interface and SQLite for the data. It also uses NumPy and pandas for calculations, Plotly and Matplotlib for charts, and Pillow to draw the puzzles. Sounds and voice go through the browser (Web Audio, speech synthesis), with no file to download.



architecture · interface, 3 blocks, local database, 2 options

The interface asks the activity banks for each question. Answers go through the engine, which updates the Brain Twin and saves them in the local database. The coach reads the engine's numbers and only uses an external AI if the parent turns it on.

File	Role
`app.py`	Interface: form, tabs, brain games, reading, writing, dashboard
`ui_extra.py`	Interactive screens: autism tools, dyslexia Build & read, quick talk board
`engine.py`	Storage, Brain Twin, levels, learning plan, profiles and degree, reading comparison
`tasks.py`	The 5 brain-game dimensions (a new dimension takes about 15 lines)
`practice.py`	Shared question bank: levels, no repetition, topic rotation, changing names
`support_content.py`, `autism_content.py`	Questions and tools of the autism profile
`focus_content.py`, `games_content.py`	ADHD profile games and shared games (robot, detective, rhythm, pause)
`literacy.py`, `dyslexia_content.py`	Writing, letter-by-letter correction, Word games, Build & read
`visual_games.py`	Puzzle and leaves game, pictures drawn by the code
`coach.py`, `collab.py`	Explainable coach and AI collaboration score
`stt.py`	Local speech recognition (optional)
`simulate.py`, `tests/`	Validation with simulated users and 71 automated tests

5. Profiles, degree and personalisation

At first launch, the parent fills in a form: first name or nickname, age, learning profile, degree of need and consent. Each profile opens its main activity first and presets comfort options, which can be changed later.

Profile	Main activities	Preset options
Typical	Brain games (5 dimensions)	Fun mode under 14
ADHD	Focus & self-control, Puzzle & leaves	Short series, stars, movement breaks

Profile	Main activities	Preset options
Autism	Feelings & routines, My tools, Puzzle & leaves	Calm mode: no timer, no animation
Dyslexia	Reading, writing, Word games, Build & read, Puzzle & leaves	Reading-friendly font (Lexend), bigger text, read-aloud

For the ADHD, autism and dyslexia profiles, the parent also chooses a degree. It only changes how the activities are presented:

Rule	Mild	Moderate	Severe
Questions per series	5	4	3
Highest level of questions and games	3	3	2
Highest brain-game level	5	4	2
Hints	on demand	already open	always shown
Default level of the tools	2	2	1

Age also limits difficulty: brain games go up to level 3 before age 8 and level 4 between 8 and 10. For questions, children aged 7 and under stay at levels 1 and 2.

6. Activities by profile

The application has more than 50 activities in three families: adaptive question games, interactive games (tiles, cards, puzzle) and tools with no right or wrong answer. Each activity has three levels and its own positive message.

ADHD — Focus & self-control tab

Series of 5 questions (depending on the degree) with stars, then a movement break ("Do 5 frog jumps").

Activity	Skill practised	Level 1 → 3
Robot rescue	Selective attention, instructions	One colour → colour and shape → the rule changes between rounds
Detective	Visual search, details	Drawn keys: 1 feature → 2 → clues from 3 scenes
Pack your school bag	Organisation	1 subject → 3 subjects (tick) → check with a list
Emotion detective	Emotions, self-regulation	Faces → a situation with several possible feelings → choose a strategy
Treasure map	Memory, planning	1 direction → 3 directions to remember → a route around rocks
Copy the rhythm	Auditory attention	2 sounds → 3-4 sounds → 5-6 sounds, answers after listening

Activity	Skill practised	Level 1 → 3
Spot the target, Same or different, Stop & go, Follow the steps, Plan & self-control	Attention, inhibition, instructions, everyday situations	Generated for every game

Autism — Feelings & routines and My tools tabs

No timer, always kind feedback, an “I need a break” button. Speaking is never required and “I don't know” is always accepted.

- **Questions (17 topics)**: feelings, faces, friends and conversation, expressions not to take literally, what others think, routines, changes of plan, body and calm, patterns, Emotion detective, My day in cards, Change detective, Conversation bubbles, Clue puzzle, Solution box, Mission: pack my bag, My interests world.
- **Tools (7)**: my day planner with changes announced in advance, emotion radar, picture talk board with voice (also in the sidebar), choice shop, my steps one at a time, sound and sensations studio (never automatic, low volume), my interests world.

Dyslexia — Reading, Writing, Word games and Build & read tabs

- **Reading**: the child reads aloud; a local model (faster-whisper) transcribes, the parent can correct it, then the voice reads the sentence and the tricky words again.
- **Writing**: look, hide, write (or copy, or dictation). Correction is word by word and letter by letter, with the type of mistake (b/d, swapped letters, missing letter…). Misspelled words become “tricky words” reused in the spelling game.
- **Word games (13 games)**: spelling, syllable train, rhyme maze, word order, sound detective, twin words, hidden letters (b/d/p/q), word market, meaning hunt, prefixes and suffixes, story in pictures, copy the rhythm, treasure map.
- **Build & read**: build the word with tiles that say their sound, sentence puzzle, magic echo (words light up while read aloud), my story in pictures.

Typical profile and shared activities

- **Brain games**: 5 dimensions with 5 levels (digit sequence, Stroop, quick arithmetic, number series, rule switching), for every profile.
- **Puzzle & leaves** (ADHD, autism, dyslexia): a puzzle of a picture drawn by the program (4, 9 then 12 pieces) and a “find the two same leaves” game, visible then as a memory game.
- **Pause station** (everyone): breathing star, floating planets or a personal pause idea, from 20 to 60 seconds.

7. Adaptive engine and no repeated questions

Each question comes from a “bank” (`practice.py`) that mixes hand-written questions and generators. A generator builds a new question at each call: new names, emojis, objects, answer order, grids or sounds.

The next question is chosen with four rules:

1. **Level**: from the last 3 answers of the activity (3 right = next level up, at most 1 right = one level down), capped by age and degree.
2. **Never seen first**: each answer is stored with the question's id, so the rule also holds from one session to the next.
3. **Topic rotation**: in “Mix” mode, the topic practised the longest time ago comes first.
4. **Oldest comes back**: when everything at a level has been seen, the question seen the longest time ago returns, with other names and another answer order.

Brain games regenerate an exercise if it matches one of the last 60. Reading and writing sentences follow rule 2. The progress bar shows ✅ for a right answer and ❌ for a mistake.

8. Brain Twin, explainable coach and collaboration score

Brain Twin

For each dimension, the application keeps a probability distribution of the child's level on a 0 to 100 scale, updated after every exercise (Bayesian method on a grid). The probability of success follows this formula:

```
P(\text{success}) = c + (1 - c) \cdot \sigma\left(\frac{s - d}{12}\right), \quad d = (\text{level} - 0.5) \times 20
```

Here c is the chance of guessing right, s the child's level and d the difficulty. A wrong answer counts 0, a right one 1, right but slow 0.85, and at most 0.7 with a hint. A forgetting factor of 0.97 gives more weight to recent exercises. The dashboard shows the estimate, the 95% interval, a radar, progress curves, a success heatmap by level and a cautious projection.

Explainable coach

The coach receives the Brain Twin numbers, the behaviour signals and the self-reported mood, and must cite these numbers to justify each piece of advice. It has three modes:

Mode	How to turn it on	Behaviour
Rules (default)	Nothing to do	Fixed summary: strongest area, next priority, tip
Ollama (free, local)	`COACH_PROVIDER=ollama`	An AI model on the computer answers the question
Anthropic API	`ANTHROPIC_API_KEY`	The Claude model, online

A filter blocks any answer containing a medical term (for example “diagnosis”, “ADHD” or “autism”) and replaces it with the fixed summary. A “Facts” line, computed by the program, keeps the numbers exact.

AI collaboration score

```
Score = 0.20 \cdot Clarification + 0.25 \cdot Verification + 0.20 \cdot Iteration + 0.20 \cdot Autonomy + 0.15 \cdot Responsible\ use
```

Verification is measured with “spot the error” challenges: the coach states something about the child's profile, and it is wrong half of the time. The child earns more points by checking their data before answering. These weights are a design proposal, not a validated standard.

9. Validation and tests

The Brain Twin recovers the true level of simulated users with a mean error of 8.4 points out of 100 after 40 exercises, better than a fixed or random level. These virtual users have a known level and do not exactly follow the application's model: a different slope and 5% careless mistakes (`simulate.py`).

Level selection	Mean error (points)	95% interval coverage
Adaptive (our method)	8.4	87%
Fixed level (3)	9.3	93%
Random level	12.0	87%

For a true level of 70, the estimate is 66 for a typical user and 3 for a user who guesses (as expected). It drops to 35 for an always-slow user and to 27 for a user who takes many hints: this is a known limitation, and calm mode removes the slowness penalty. The interval is slightly over-confident (87% instead of 95%).

71 automated tests pass (`pytest -q`). They check that every generated question is valid, that none repeats over 30 draws, that the letter-by-letter correction is right and that sounds never start on their own. Some tests run the application in memory and play like a child: create a profile with a degree, finish an ADHD series up to the movement break, build a word, complete a puzzle. Every new screen was also opened in a real browser (Chromium), without errors.

10. Limitations and future work

The main limitation is that no study has yet been done with real children and specialists: the effectiveness of the activities still has to be shown.

Current limitations:

- The difficulty levels are designed, not clinically calibrated; validation uses simulated users.
- The content is in English, with short sentences; the keywords of the collaboration score are in English only.
- Speech recognition is less accurate on children's voices and accents, which is why the parent can correct the transcript.
- Speed and hints lower the estimate: a slow child is at a disadvantage outside calm mode.
- A generated question can sometimes offer a real word as a wrong answer.
- The puzzle uses two buttons (take, place) because Streamlit cannot receive drag-and-drop.

Future work:

1. Test the application with families and with speech therapists, psychologists or special-needs teachers.
2. Translate the interface and content into French and Arabic.

3. Let the parent add their own photos and pictograms (day planner, talk board).
4. Improve the rule-based coach so it answers simple questions without AI.
5. Recalibrate the levels from real data, with the families' consent.

11. Installation and use

The application needs **64-bit** Python 3.11: with 32-bit Python, the pyarrow library required by Streamlit cannot be installed. On Windows (PowerShell), in the project folder:

1. Create the environment: `py -3.11 -m venv venv`
2. Activate it: `venv\Scripts\activate`
3. Install the libraries: `pip install -r requirements.txt` (optional voice: `pip install -r requirements-voice.txt`)
4. Start the application: `streamlit run app.py`, which opens at http://localhost:8501
5. Run the tests: `pytest -q`

For a coach that really understands questions, install Ollama, run `ollama pull llama3.2:3b`, then `$env:COACH_PROVIDER="ollama"` before `streamlit run app.py`. This setting only lasts for the open PowerShell window.

Typical use: the parent creates the profile, the child starts with their profile's main tab, then the parent checks the dashboard (estimates, progress, tricky words, breaks taken).

12. Conclusion

Cognitive Growth Agent brings together, in one local application, more than 50 activities adapted to ADHD, autism and dyslexia, progress tracking that shows its uncertainty and a coach that explains its advice. The profile and degree, chosen by the parent, set the content, series length, difficulty and help, without ever making a diagnosis.

The project works and is tested (71 automated tests, checks in the browser). The essential next step is an evaluation with children, families and specialists, to measure the real effect of the activities and adjust the levels.